

DATA VISUALISATION

Unit-1

What Is Visualization?

We define visualization as the communication of information using graphical representations. Pictures have been used as a mechanism for communication since before the formalization of written language. A single picture can contain a wealth of information, and can be processed much more quickly than a comparable page of words.

Eg. 1) • a table in a newspaper, representing data being discussed in an article;

- 2) a weather chart showing the movement of a storm
- 3) a graph of stock market activities
- 4) 3D reconstruction of your injured knee,

Visualisation is important

Visualization makes data easy to understand by turning it into pictures and graphs. It helps people see important things in data, make better choices, and find things that are unusual or wrong. It's like using pictures to tell a story about data.

Visualization most often provides different levels of both qualitative and quantitative views of the information being communicated.

Visualization provides a visual representation of objects that may include data, algorithms, results of computations, processes, user controls, and numerous other components of an application. These visual representations provide information through the use of computer-generated graphics

Relationship between Visualization and Other Fields

Aspect	Visualization	Computer Graphics
Definition	Utilizes graphics to display data and information.	Focuses on creating interactive synthetic images and animations with an emphasis on visual realism.
Use of Graphics	Graphics are used as a communication medium.	Primarily focused on creating graphical objects and achieving visual realism, often in art and entertainment.
Data Focus	Based on underlying data, including spatial positions, populations, or physical measures.	Focuses primarily on graphical objects and the organization of graphic primitives.
Interdisciplinary	Encompasses aspects from various disciplines, including human-computer interaction, perceptual psychology, databases, statistics, and data mining.	Less interdisciplinary, with a primary focus on creating visual content.
Goal	Emphasizes effective communication of information rather than visual realism.	Often aims to achieve visual realism as a primary goal.
Data Types	Deals with various types of data, which may not involve physical objects, focusing on attributes that may not be visible.	Typically deals with the display of physical objects or synthetic images.

Aspect	Scientific Data Visualization	Information Visualization
Data Types	Primarily deals with scientific and	Focuses on representing various types
	numerical data, often from physical	of data, including non-scientific or
	experiments, simulations, or observations.	non-numerical data, such as text,
		social trends, or categorical data.
Data Characteristics	Often involves complex and structured	Frequently deals with diverse and
	data sets, such as 3D models, medical	unstructured data, which may require
	images, or simulations with high	cleaning, transformation, or aggregation.
	spatial or temporal dimensions.	
Visualization Goals	Aims to enable in-depth analysis and	Focuses on making data more
	understanding of scientific phenomena,	accessible and understandable to a wide
	facilitating scientific discovery and	audience, emphasizing in

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	insights within the domain of science.	patterns in the data.
Computational Complexity	Often involves computationally intensive	Generally involves less computational
	tasks due to complex data and the need	complexity, with a focus on data
	for data processing and simulation.	presentation and exploration.
Examples	Examples include visualizing the	Examples include creating charts,
	molecular structure of proteins,	graphs, scatterplots, word clouds,

"The Visualization Process":

1. Initial Analysis: The visualization process starts with an analysis of the type of data available and the information the viewer wants to extract or convey with the display.
2. Data Sources: Data can come from various sources and may have different levels of complexity.
3. Viewer's Intent: The viewer's purpose for using visualization can be exploration, hypothesis confirmation, or presentation of analysis results.
4. Visualization Mapping: To visualize data, a mapping from the data to the display must be defined, using various graphical objects and attributes.
5. Interactive Controls: Interactive controls are crucial in modern visualization, allowing users to control data selection, mapping, colors, and view refinement.
6. Customization and Refinement: Users often have different backgrounds and tasks, so enabling them to customize and refine visualizations is essential.
7. Part of a Larger Process: Visualization is often part of a larger process, such as exploratory data analysis, knowledge discovery, or visual analytics.

8. Data Preparation: Data preparation is important and may involve cleaning and massaging data, especially when dealing with erroneous or noisy data.

9. Building Models: The goal of visualization in data exploration is to convey information, discover new knowledge, identify patterns, anomalies, trends, and relationships, and ultimately build models that represent or approximate the data.

10. Visualization Pipeline: The process of going from data to an image or model via a computer is described as a pipeline with stages that can be studied independently.

computer graphics pipeline

1. Modeling : This is where we create a 3D model of objects by defining their shape and appearance.

2. Viewing: Imagine placing a virtual camera to look at our 3D model. This step makes sure everything looks right from the camera's perspective.

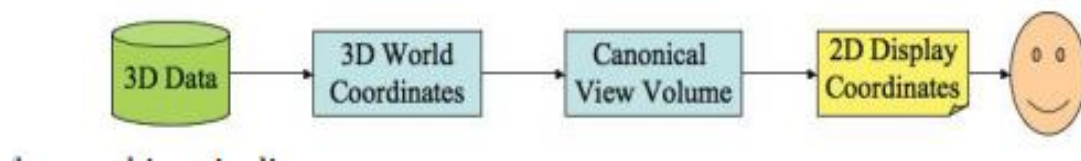
3. Clipping: Think of this like cutting out a picture. We remove things that the camera can't see, like parts of objects that are out of view.

4. Hidden Surface Removal: Here, we get rid of objects that are hidden behind others or are facing away from the camera.

5. Projection: We take our 3D model and flatten it onto a 2D plane, like when you take a photo.

6. Rendering: This step is like coloring in a coloring book. We decide the colors of the pixels on the 2D image based on things like the object's material, the lighting, and reflections.

7. Ray Tracing: Instead of following the usual steps, ray tracing is like pretending each pixel on the image is a little camera. It tracks how light travels, reflects, and interacts with objects to create very realistic images. It's a bit more complicated, though, because it has to do a lot of calculations to get it right.



Visualization Pipeline:

1. Data Modeling: Before we can make a visual representation of data, we need to organize the data. Think of it like labeling each piece of information with a name, type, and what it means, so the computer can use it easily.

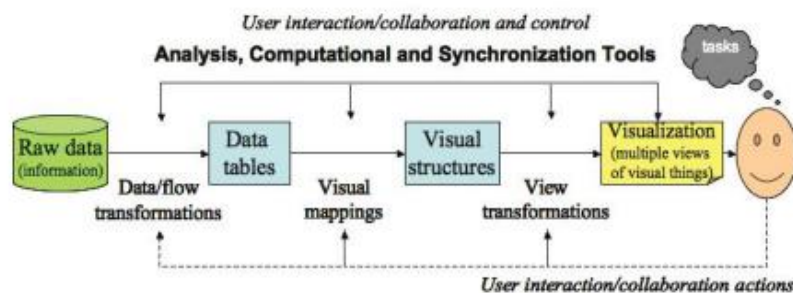
2. Data Selection: Just like choosing which part of a photo you want to show, here we decide which data to visualize. It can be chosen by the user or by the computer, based on what's important.

3. Data to Visual Mappings: This step is about connecting data with what you see on the screen. For example, data can decide the size,

position, or color of something you're looking at. Sometimes, we need to change the data a bit before it's shown, like making it bigger or changing its color.

4. Scene Parameter Setting: Similar to picking colors for a drawing, here we choose things like colors, sounds, and how light should work to make the visualization look right. It's like setting the stage for the data to shine.

5. Rendering or Generation: This is where the computer draws what you see on the screen. It might involve shading, textures, or just drawing lines and shapes. The final visualization often includes extra information like labels and keys to help understand the data.



Knowledge Discovery (Data Mining) Pipeline:

1. Data: Just like in other pipelines, it all begins with the data. In data mining, data is a big deal, and we focus on it more than in graphics and visualization.

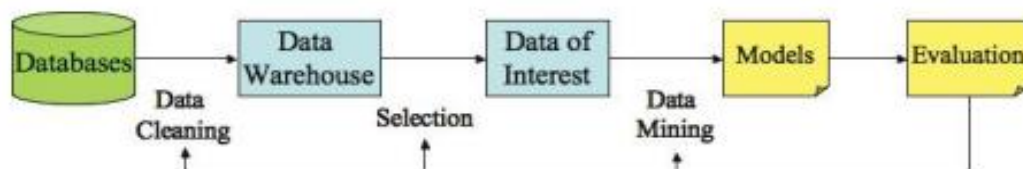
2. Data Preparation: Before we start analyzing data, we have to get it ready. This step involves cleaning, organizing, and choosing the right data sets for analysis.

3. Data Mining: This is the core of data mining. We use algorithms to dig deep into the data, looking for patterns and creating a model to understand it better.

4. Pattern Evaluation: We check how good our model is. Is it stable, accurate, and reliable? This step helps us make sure our findings are solid.

5. Rendering or Visualization: After finding useful patterns, we need to show them to people. It's like creating a graph or chart to help others understand what we discovered.

6. Interactive Visualization: At any point, we can use interactive visual tools to help with the whole process. It's like steering a ship to the right direction while sailing.



the role of perception in data visualization:

1. Human Visual System: When we create visualizations, we need to think about how people see things. We want to make sure our visualizations are clear and don't create confusion. For example, we don't want to show something that looks like it's moving when it's not.

2. Preattentive Processing: Our eyes quickly notice differences in things like color or texture. For example, you can easily spot a differently colored dot in a group. This helps in making quick comparisons.

3. Key Perceptual Elements: We also notice things like the orientation of lines, the size of objects, and motion. All of this happens almost instantly without us really thinking about it.

4. Guidelines for Visualization: Understanding how our brain sees things gives us guidelines for making visualizations. For example, we know that certain colors are hard to distinguish for some people, so we should allow users to customize colors. We also use principles from psychology, like how people naturally group things together or see patterns, to make visualizations easier to understand.

Pseudocode conventions:

1. Data: In pseudocode, we assume the data is like a table with numbers. If your data has non-numeric parts, you need to change them into numbers. When you work with a part of the data, you pretend it's a smaller table.

2. Number of Dimensions and Records: Dimensions are like columns in the table, and records are like rows. We use "m" for the number of columns and "n" for the number of rows.

3. Normalize: This is a way to change the data so that it's easier to work with. It's like making all the numbers fit between a certain range. You can do this for each column, and sometimes you can

choose your own range. For example, you might want all the numbers to be between 0 and 1.

4. Color: This is about setting the colors you'll use for drawing in your program. You can set a color using an RGB value (a way to describe colors with numbers).

5. MapColor: This is like picking a color based on the data. For example, you might want to make the color of a shape depend on a number in your data.

6. Circle: This is how you draw a filled circle on your screen. You say where the center is and how big it is. In this case, it's all done on a unit square (a square with sides of length 1).

7. Polyline: This helps you draw a bunch of connected lines, like drawing a shape by connecting dots.

8. Polygon: This is used to fill a shape with a color, like coloring inside the lines of a coloring book.

Scatterplots

Scatterplots are a fundamental data visualization technique used to display data points in a two-dimensional space. They are essential for understanding data patterns and relationships between variables. Here are the key characteristics of scatterplots:

- Data Representation: Data points are represented as circles in a scatterplot. These circles are positioned on the plot to correspond to values from two different dimensions, typically referred to as `xDim` and `yDim`.

Key Benefits of Scatterplots

Scatterplots offer several advantages for data visualization:

1. Data Understanding: They enable quick data comprehension by visually representing data relationships and patterns.
2. Hypothesis Generation: By observing the positions and patterns of data points, users can generate hypotheses about the data.
3. Pattern Identification: Scatterplots help identify trends, correlations, and patterns within the data.

In summary, scatterplots are a valuable tool in data analysis, allowing users to visually explore data relationships and make informed observations about the dataset.

The role of the user in visualization can vary based on the purpose of the visualization:

1. Exploration: Users examine a data set to understand its contents and identify specific features.

2. Confirmation: Users verify the presence of certain features in the data, often based on prior analysis or hypotheses.

3. Presentation: Users create visualizations to communicate concepts or facts to an audience, often using labeling and strong colors for emphasis.

4. Interactive Presentation: Users deliver interactive presentations that allow individuals to explore the data. These presentations are

DATA FOUNDATIONS

- Data is the starting point for visualization and can come from various sources, including sensors, surveys, simulations, and computations.
- Data can be either raw or processed, involving techniques like smoothing, scaling, noise removal, or interpolation.
- A typical data set used in visualization contains n records, where each record (r_i) comprises m observations or variables.
- Variables can be classified as independent (iv_i) or dependent (dv_j), where independent variables are unaffected by other variables, while dependent variables are influenced by them.
- Records can be represented as $r_i = (iv_1, iv_2, \dots, iv_{m_i}, dv_1, dv_2, \dots, dv_{m_d})$, where m_i is the count of independent variables, and m_d is the count of dependent variables.
- Independent variables are often seen as the domain of a function, and dependent variables as the range.
- Data sets do not necessarily encompass all possible combinations of values within their domain, and the distinction between dependent and independent variables can be unclear in some cases.

Types of Data

In its simplest form, each observation or variable of a data record represents a single piece of information. We can categorize this information as being ordinal (numeric) or nominal (nonnumeric). Subcategories of each can be readily defined.

Ordinal. The data take on numeric values:

- binary—assuming only values of 0 and 1;
- discrete—taking on only integer values or from a specific subset (e.g., (2, 4, 6));
- continuous—representing real values (e.g., in the interval [0, 5]).

Nominal. The data take on nonnumeric values:

- categorical—a value selected from a finite (often short) list of possibilities (e.g., red, blue, green);
- ranked—a categorical variable that has an implied ordering (e.g., small, medium, large);
- arbitrary—a variable with a potentially infinite range of values with no implied ordering (e.g., addresses).

Scale. Three attributes that define a variable's measure are as follows:

- Ordering relation, with which the data can be ordered in some fashion. By definition, ranked nominal variables and all ordinal variables exhibit this relation.
- Distance metric, with which the distances can be computed between different records. This measure is clearly present in all ordinal variables, but is generally not found in nominal variables.
- Existence of absolute zero, in which variables may have a fixed

lowest value. This is useful for differentiating types of ordinal variables. A variable such as weight possesses an absolute zero, while bank balance does not.

Structure within and between Records

Data sets exhibit structure in terms of how they are represented (syntax) and the relationships within records and between records (semantics).

Scalar ,vector and tensors

1. Scalars: Single data values, often the focus for analysis and visualization. Examples include item cost and individual age.
2. Vectors: Combinations of variables. For instance, a 2D flow field point can be represented by displacement values in x and y.
3. Tensors: General structures defined by rank and dimensionality. Scalars are rank 0 tensors, while vectors are rank 1 tensors. Tensors of higher ranks can be represented as arrays or matrices. An example is a transformation matrix for specifying a local coordinate system in a data record.

Geometry and grids

- Geometric Structure: Common in scientific and engineering datasets.
- Explicit Coordinates: Data records include precise coordinates (e.g., longitude and latitude).
- Implied Structure: Assumes data records are positioned on a grid.

- Coordinate Systems: Various systems used (e.g., Cartesian, spherical).
- Complexity with Nonuniform Geometry: Transition to nonuniform grids increases computational complexity.

Other structure

- **Timestamps:** Vital attribute with a wide time range, from picoseconds to millennia, relative or absolute.
- **Temporal Attributes:** Can be uniformly spaced (e.g., continuous phenomenon) or nonuniformly spaced (e.g., business transactions).
- **Topology and Connectivity:** Data records have relationships (topology), which is essential for processes like resampling and interpolation

Examples of various structured data

Certainly, here are those examples of structured data described in an organized manner:

1. MRI (Magnetic Resonance Imagery) Data:

- Data Type: Density (Scalar)
- Attributes: Three spatial attributes
- Structure: 3D grid connectivity

2. CFD (Computational Fluid Dynamics) Data:

- Data Type: Three dimensions for displacement
- Attributes: One temporal and three spatial attributes
- Structure; 3D grid connectivity (can be uniform or nonuniform)

3. Financial Data:

- Data Type: No geometric structure
- Attributes: Possibly n independent components
- Attribute Types: Nominal and ordinal
- Structure: Includes a temporal attribute

4. CAD (Computer-Aided Design) Data:

- Attributes: Three spatial attributes
- Structure: Includes edge and polygon connections, as well as surface properties

5. Remote Sensing Data:

- Attributes: Multiple channels
- Attribute Types: Two or three spatial attributes
- Structure: Includes one temporal attribute and grid connectivity

Data Preprocessing:

- In most cases, it's preferable to work with the original raw data.
- Data preprocessing methods are used to prepare the data for visualization and analysis, but their specific application depends on the context and objectives of the analysis.

Data preprocessing involves various techniques such as normalization, segmentation, sampling, dimension reduction, mapping nominal dimensions, aggregation, smoothing, and raster-to-vector conversion. These processes enhance visualization and may lead to new insights. It's crucial to communicate the applied transformations to users to ensure correct data interpretation and prevent misinterpretation.

Common methods/technique for preprocessing

1) metadata and statistics:

- Metadata, which includes information about the data set, is essential for interpreting the data. It can include details on field formats, reference points, units of measurement, missing value indicators, and measurement resolution. This information helps in selecting appropriate preprocessing operations.
- Statistical analysis methods can provide valuable insights:
 - Outlier detection helps identify records with erroneous data.
 - Cluster analysis can group similar data points.
 - Correlation analysis reveals relationships between dimensions.
- Common statistics for data analysis include the mean (μ) and standard deviation (σ).
- Histograms are commonly used to visualize the distribution of data.

2) dealing with missing values and data cleaning:

1. Reasons for Missing Data: Missing data can result from various causes, including malfunctioning sensors, blank survey entries, or human errors.

2. Strategies for Handling Missing Data: Common strategies include discarding records with missing or erroneous fields, assigning sentinel values to mark problematic data, or substituting missing values with the average, nearest neighbor values, or computed substitute values. Each strategy has its advantages and limitations.

3. normalization

1. Normalization Purpose: Data normalization is a crucial step in data preprocessing, aiming to standardize data so that it satisfies specific statistical properties. This transformation allows for meaningful comparisons and effective visualization.

2. Common Techniques: Common normalization techniques involve scaling data to fit within a common range, ensuring uniform mean and standard deviation, or applying non-linear transformations. This adjustment is essential to match data with the range of graphical attributes used in visualization.

$$d_{\text{normalized}} = (d_{\text{original}} - d_{\text{min}}) / (d_{\text{max}} - d_{\text{min}}).$$

$$d_{\text{sqrt-normalized}} = (\sqrt{d_{\text{original}}} - \sqrt{d_{\text{min}}}) / (\sqrt{d_{\text{max}}} - \sqrt{d_{\text{min}}})$$

4) Sampling and Subsetting:

These techniques involve transforming a dataset to a different spatial resolution or selecting specific portions of data for visualization. Sampling methods include linear and nonlinear interpolation for estimating values at unsampled locations. Subsetting involves filtering or selecting data based on specific criteria, like time periods or data attributes. These operations are essential for managing large datasets and controlling visual clutter.

5) Dimension Reduction:

In situations where the data has too many dimensions for effective visualization or analysis, dimension reduction techniques are used. These methods, like Principal Component Analysis (PCA), Multidimensional Scaling (MDS), Self-organizing maps (SOMs), and

Local Linear Embedding (LLE), transform high-dimensional data into a lower-dimensional space while preserving significant features and patterns. This process makes complex data more manageable and interpretable.

6) Mapping Nominal to Numeric:

Converting non-numeric categorical data to numbers for visualization. Use labels, colors, or shapes to distinguish values or analyze statistical properties for meaningful mappings. Involves techniques like MDS for multi-nominal variables.

7) Smoothing and Filtering:

Techniques like convolution and exponential smoothing are employed to reduce noise and soften discontinuities in data. Convolution involves weighted averaging of neighboring data points, while exponential smoothing weights more recent data more heavily in time series analysis.

8) Raster-to-Vector Conversion: In spatial data visualization, this process involves transforming raster-based data, like images, into vertex and edge-based models. Techniques for this conversion include thresholding, region-growing, boundary detection, and thinning, enabling tasks like data compression, comparison, transformation, and segmentation.

Dimension Reduction

Visualization can process limited dimensions only.

Reduce dimension but preserve the information contained.

Can be done manually
PCA, Multidimension Scaling, SOM etc.

PCA

Set of Correlated variables to set of uncorrelated variables.

also a unsupervised learning algo.
reduce dimensionality but preserve variables.

- ① Standardize the range of continuous initial variables
- ② Compute Covariance matrix to find Correlation
- ③ Compute eigenvectors + eigenvalues of Covariance matrix to find principle components
- ④ Create a feature vector. decide what to keep.
- ⑤ recast the data along principle component axes.

① Standardize (same scale)

$$Z = \frac{\text{Value} - \text{mean}}{\text{stdev}}$$

② Covariance matrix

variables varying ~~to~~ from mean
wrt to each other

$P \times P$ matrix, P ; P = num of dimen

x, y, z variables

$\text{cov}(x, x)$	(x, y)	(x, z)
(y, x)	(y, y)	(y, z)
(z, x)	(z, y)	(z, z)

Smoothing (convolution)

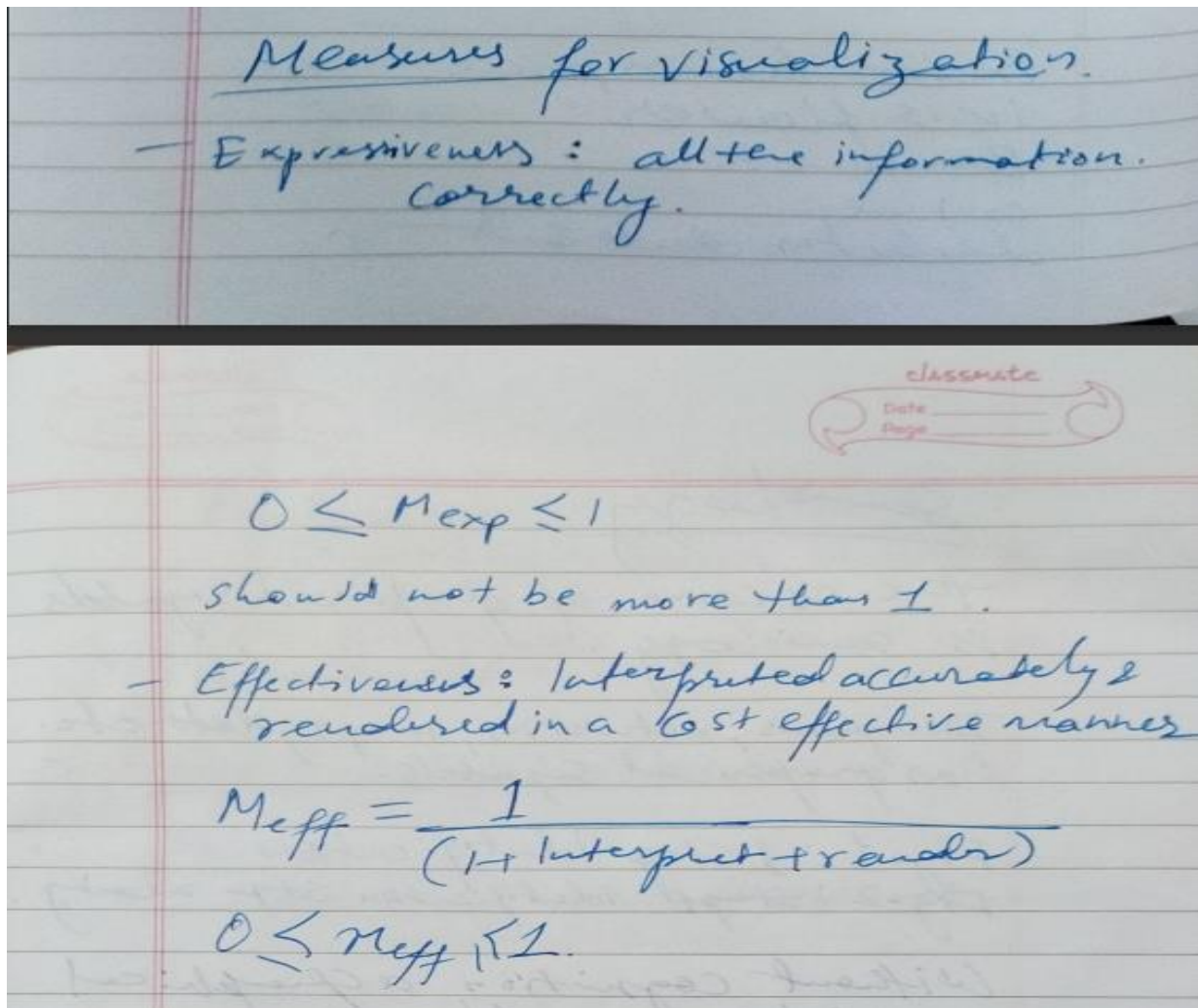
DATASETS

characteristics of a good dataset

1. high accuracy & units etc.
2. Reliability: no contradictions.
 - a) duplicate values
 - b) missed values
 - c) special chars etc.
 - d) ~~etc~~
3. Consistency in feature representation
 - fixed size images.
 - numeric values.
 - normalized values.
4. size of data set
5. Diversity
6. Completeness
7. Relevance
8. updated.

Unit-2

visualization



- **Semiology of Graphical Symbols:** Semiology is the science of designing and perceiving graphical symbols and marks. It plays a crucial role in creating effective visualizations, ensuring that symbols are well-designed, easily recognizable, and convey meaningful information.

- **Symbols and Visualizations:** Symbols can be preattentively recognizable (easily understood) or may require more attention to interpret. The effectiveness of a symbol lies in its ability to convey meaning quickly and efficiently.

- Features of Graphics: Graphics have multiple dimensions, often represented in 2D or 3D views. These dimensions correspond to the data's variables, and any patterns in the graphic should reflect patterns in the data. Graphics can represent a large number of data items, and the eye can perceive distributions and relationships within the graphic.

- Analysis of a Graphic: Analyzing a graphic involves perceiving groups of objects, characterizing these groups, and examining special cases or relationships between groups. Supporting analysis techniques, such as clustering, can aid in data interpretation and group perception.

The Eight Visual Variables are fundamental attributes used in data visualization to encode and convey information:

1. Position: The arrangement of graphical objects in a display space, providing the fundamental foundation for reading a visualization.
2. Mark (Shape): The shape of graphical marks, such as points or symbols, used to represent different data values.
3. Size (Length, Area, Volume): The size of marks, which can indicate the magnitude of data values.
4. Brightness: The degree of brightness used to distinguish marks and encode relative differences.
5. Color: The use of hue and saturation to represent data values with colors.
6. Orientation: The direction or rotation of marks to convey data relationships.
7. Texture: The use of patterns or textures created by marks, colors, and orientation to add data-related attributes.

8. Motion: The use of dynamic changes over time to convey evolving data patterns or trends.

Effects of visual variables

1. Selective Visual Variables: Used to spontaneously divide data values into distinguished groups (e.g., for nominal values).

- Effective visual variables: Size, Brightness, Texture, Color (primary colors), Direction/Orientation.

2. Associative Visual Variables: All factors have the same visibility, suitable for visualizing nominal values.

- Effective visual variables: Texture, Color, Direction/Orientation, Shape.

3. Ordinal Visual Variables: Create spontaneous order among data values, suitable for visualizing ordinal and quantitative data.

- Effective visual variables: Texture, Size, Brightness.

4. Proportional Visual Variables: Establish a direct association of relative size, suitable for visualizing ordinal and quantitative data.

- Effective visual variables: Size, Direction/Orientation, Brightness.

5. Separating Visual Variables: Ensure all elements are visible while making the rest not visible.

- Effective visual variables: Texture, Color, Direction/Orientation, Shape.

Taxonomies

A taxonomy is a means to convey a classification. Often hierarchical in nature, a taxonomy can be used to group similar objects and define relationships.

In the context of visualization taxonomies, here is a brief summary of the taxonomies mentioned:

1. Keller and Keller (1994) Taxonomy of Visualization Goals:

- Classifies visualization techniques based on data types and user tasks.
- Data types include scalar, nominal, direction, shape, position, and spatially extended region or object.
- User tasks consist of identifying, locating, distinguishing, categorizing, clustering, ranking, comparing, associating, and correlating.

2. Shneiderman (1996) Data Type by Task Taxonomy:

- Categorizes data types as one-dimensional linear, two-dimensional map, three-dimensional world, temporal, multidimensional, tree, and network.
- Focuses on the behavior of analysts and their tasks, including overview, zoom, filter, details-on-demand, relate, history, and extract.

3. Keim (2002) Information Visualization Classification:

- Classifies visualization systems based on three dimensions: data types, visualization techniques, and interaction/distortion methods.
- Data types encompass one-dimensional, two-dimensional, multidimensional, text and hypertext, hierarchies and graphs, and algorithm and software.

- Visualization techniques are divided into standard 2D/3D displays, geometrically transformed displays, iconic displays, dense pixel displays, and stacked displays.

- Interaction/distortion techniques include dynamic projection, interactive filtering, interactive zooming, interactive distortion, and interactive linking and brushing.

These taxonomies help categorize and understand various aspects of visualization, including data types, techniques, and interaction methods.

Historical perspective of data visualization:

Bertin (1967) Semiology of Graphics:

1. Bertin emphasized the separation of content from container, providing a fundamental concept in visualization.
2. He introduced a graphical vocabulary that included plane properties and retinal variables, which are used for encoding information.

Mackinlay (1986) APT:

1. Mackinlay introduced the concept of graphical languages and defined criteria for expressiveness and effectiveness.
2. He developed a composition algebra for creating complex presentations by merging primitive graphical languages.

Bergeron and Grinstein (1989) Visualization Reference Model:

1. The Visualization Reference Model established a conceptual visualization pipeline with four stages to map data to visual representations.
2. It served as a framework for guiding the visualization process, from data source to perceptual stimulation.

Wehrend and Lewis (1990):

1. Wehrend and Lewis created a catalog of encoding techniques and their effective uses, which could serve as a knowledge base for visualization.
2. The catalog was designed to match domain tasks and goals with appropriate visualization techniques, although it was not implemented in an automated system.

Robertson (1990) Natural Scene Paradigm:

1. Robertson aimed to leverage human skills in analyzing natural scenes for multivariate data analysis.
2. He defined visual properties of natural scenes and matched them with data variables, prioritizing interpretation aims for selecting appropriate representations.

Gibson's Affordance Theory, developed by psychologist James J. Gibson, is a concept in perceptual psychology and ecological psychology. It focuses on how organisms, particularly humans, perceive and interact with their environment based on the information provided by the environment itself. According to this theory, the environment offers various "affordances" or action possibilities, and individuals perceive these affordances and respond to them

accordingly. It's a way of understanding how perception is inherently linked to action.

Here are five examples of semiotics (the study of signs and symbols) based on Gibson's Affordance Theory:

1. Door Handle: In an environment, a door handle affords the action of grasping and turning. When you see a door handle, it serves as a sign that you can use it to open the door.

2. Chair: A chair affords the action of sitting. When you see a chair, you interpret it as a sign that you can sit on it.

3. Stairs: Stairs afford the action of ascending or descending. When you encounter a staircase, it signals to you that you can use it to move between different levels of a building.

4. Button: A button on an electronic device affords the action of pushing. It serves as a sign indicating that you can press it to perform a specific function.

5. Pedestrian Crossing Sign: A road sign depicting a pedestrian crossing serves as a semiotic affordance. It communicates to drivers that pedestrians might cross the road at that location, indicating the need to be cautious.

These examples demonstrate how the concept of affordances in Gibson's theory relates to semiotics. Affordances are the inherent qualities of objects or elements in the environment that provide

meaningful information and guide individuals' actions and interpretations.